Errata File for Second Edition

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This errata file contains a list of errors, short clarifications and modifications to the manuscript's second edition.

1. Page 7, first sentence below equation (1.5): "This means that there is a 95% probability that this interval contains the true parameter θ".

   This statement is technically incorrect for the following reason. A $100\,(1-\alpha)\,\%$ confidence interval for θ is an interval defined by the lower and upper bounds $(l\,(Y)\,, u\,(Y))$, where Y represents the data, a random variable, such that

   $$\Pr\left[l\,(Y) < \theta < u\,(Y)\,|\theta\right] = 1-\alpha,$$

   where θ is the fixed, unknown parameter and α is the significance level of the test. The probability is computed with respect to the distribution of Y. Once the data are observed and $Y = y$, the expression

   $$\Pr\left[l\,(y) < \theta < u\,(y)\,|\theta\right] = 1-\alpha,$$

   does not make sense since there are no random variables involved within the square brackets. It is therefore technically incorrect to assert that the observed interval $(l\,(y)\,, u\,(y))$ contains θ with probability $(1-\alpha)$. The interval $(l\,(y)\,, u\,(y))$ either contains or does not contain θ. However, if $\alpha = 0.05$, one can be quite confident that the unobserved θ is within the observed $(l\,(y)\,, u\,(y))$ and act accordingly.

   The remaining statements in the paragraph are correct.

2. Page 170, bottom half of the page, bracket missing:
   CHANGE:

   $$\mathrm{Var}\,(\hat{a}-a) = \mathrm{E}_y(\mathrm{Var}(a|y)$$

   TO.

   $$\mathrm{Var}\,(\hat{a}-a) = \mathrm{E}_y\,(\mathrm{Var}(a|y))$$

3. Page 171, the first partial derivative below Eq. (4.62).
   CHANGE:

$$\frac{\partial \ln p\left(a,b|y,b_0,a_0,B,G,R\right)}{\partial b} = -B^{-1}(b-b_0) + XR^{-1}\left(y - Xb - Za\right)$$

   TO:

$$\frac{\partial \ln p\left(a,b|y,b_0,a_0,B,G,R\right)}{\partial b} = -B^{-1}(b-b_0) + X'R^{-1}\left(y - Xb - Za\right)$$

4. Page 172, top.
   CHANGE: "When B^{-1} implying $B = \infty$ ..."
   TO: "When and if $B^{-1} = 0$ implying $B = \infty$ ..."

5. Page 172,
   CHANGE:
   Finally, add and subtract $\hat{\theta}'\left(W'R^{-1}W + \Sigma\right)\hat{\theta}$ and rearrange to yield

$$y'R^{-1}y + \left(\theta - \widehat{\theta}\right)'\left(W'R^{-1}W + \Sigma\right)\left(\theta - \widehat{\theta}\right)' - \widehat{\theta}'\left(W'R^{-1}W + \Sigma\right)\widehat{\theta} + \theta_0'\Sigma\theta_0$$

   TO:
   Finally, add and subtract $\hat{\theta}'\left(W'R^{-1}W + \Sigma\right)\hat{\theta}$ and rearrange to yield

$$y'R^{-1}y + \left(\theta - \widehat{\theta}\right)'\left(W'R^{-1}W + \Sigma\right)\left(\theta - \widehat{\theta}\right) - \widehat{\theta}'\left(W'R^{-1}W + \Sigma\right)\widehat{\theta} + \theta_0'\Sigma\theta_0$$

6. Page 172, equation at the bottom,
   CHANGE:

$$\ln p\left(\theta|y,\theta_0,B,G,R\right) \propto -\frac{1}{2}\left(\theta - \widehat{\theta}\right)'\left(W'R^{-1}W + \Sigma\right)\left(\theta - \widehat{\theta}\right)'$$

   TO:

$$\ln p\left(\theta|y,\theta_0,B,G,R\right) \propto -\frac{1}{2}\left(\theta - \widehat{\theta}\right)'\left(W'R^{-1}W + \Sigma\right)\left(\theta - \widehat{\theta}\right)$$

7. Page 263,
   CHANGE,

$$v_\beta^* = v + 1,$$

   TO:

$$v_\beta^* = v_\beta + 1,$$

8. Page 265,
   Delete Subsection "An Alternative Parametrisation" (up to and including eqn (5.82)).

9. Page 353, Eq. (7.41), remove "comma" before "$i = 1, \ldots, m$"
   CHANGE:

$$y|\mu, \alpha, \delta, \sigma^2 \sim N\left(1\mu + \sum\nolimits_{i=1}^{m} X_i\left(\alpha_i\delta_i\right), I\sigma^2\right), \quad , i = 1, \ldots, m \tag{7.41}$$

   TO:

$$y|\mu, \alpha, \delta, \sigma^2 \sim N\left(1\mu + \sum\nolimits_{i=1}^{m} X_i\left(\alpha_i\delta_i\right), I\sigma^2\right), \quad i = 1, \ldots, m \tag{7.41}$$

10. Page 355, CHANGE:
    Let $(y - 1\mu - X_{-i}b_{-i})'(y - 1\mu - X_{-i}b_{-i}) = RSS0$, the residual sum of squares of the model without the ith SNP and let $(y - 1\mu - Xb)'(y - 1\mu - Xb) = RSS1$ denote the residual sum of squares of the model with all the m SNP's.
    TO:
    Let $(y - 1\mu - X_{-i}b_{-i})'(y - 1\mu - X_{-i}b_{-i}) = RSS0$, the residual sum of squares of the model with the SNP i excluded and let $(y - 1\mu - Xb)'(y - 1\mu - Xb) = RSS1$ denote the residual sum of squares of the model with the SNP i included.

11. Page 356, REPLACE THE CONTENTS OF THE SUBSUBSECTION:
    " **Remarks**"
    WITH:
    It will be convenient to write $D = (H, y)$, where H represents the values of all unknown parameters of the model other than δ_i and y denotes the observed data.

    (a) Residuals $RSS0$ and $RSS1$ are efficiently updated as indicated on page 344.

    (b) In a computing environment (7.53) and (7.54) can be obtained as follows. Let $\theta_i(H) = \Pr(\delta_i = 1|H, y)$ denote the posterior probability that marker i is non-zero, conditional on parameters H. Then

$$\ln\left(\frac{\theta_i(H)}{1 - \theta_i(H)}\right) = \frac{1}{2\sigma^2}\left(RSS0 - RSS1\right) - \left(\ln\left(1 - \pi\right) - \ln\left(\pi\right)\right) = K_i \tag{7.56}$$

    and

$$\theta_i(H) = \frac{\exp\left(K_i\right)}{1 + \exp\left(K_i\right)}. \tag{7.57}$$

    For marker i, let $\theta_i^{[j]} = \theta_i(H^{[j]}) = \Pr(\delta_i = 1|y, H^{[j]})$ be the computed value of $\theta_i(H)$ at round j of an McMC implementation. The quantity $\theta_i^{[j]}$ is not itself an additional draw conditional on $H^{[j]}$. Once $H^{[j]}$ has been drawn, $\theta_i^{[j]}$ is calculated deterministically.

Since $\delta_i|y, H^{[j]} \sim \text{Bernoulli}(\theta_i^{[j]})$, $\text{E}(\delta_i|y, H^{[j]}) = \theta_i^{[j]}$. Further, marginally over the distribution of $H|y$,

$$\begin{aligned}\Pr(\delta_i = 1|y) &= \int \Pr(\delta_i = 1|y, H)p(H|y)dH \\ &= \text{E}_H(\theta_i(H)|y).\end{aligned}$$

This expectation can be approximated using the Monte Carlo realisations $\theta_i^{[j]}$, $j = 1, 2, \ldots, T$

$$\hat{\text{E}}_H(\theta_i(H)) = \frac{1}{T}\sum_{j=1}^{T}\theta_i^{[j]}. \tag{7.58}$$

Therefore, the algorithm provides two characterisations of the marginal posterior inclusion probability $\Pr(\delta_i = 1|y)$: one based on the mean of sample indicators $\delta_i^{[j]}$ (7.55), and the other given in (7.58), based on the mean of the inclusion probabilities $\theta_i^{[j]}$. This second estimator is a Rao-Blackwell-type estimator: it replaces each sampled Bernoulli indicator $\delta_i^{[j]}$ by its conditional expectation $\theta_i(H^{[j]}) = \text{E}(\delta_i|H^{[j]}, y)$. Therefore it eliminates the additional Bernoulli sampling variation $\theta_i(H^{[j]})(1-\theta_i(H^{[j]}))$ associated with drawing $\delta_i^{[j]}$ while retaining the variation in $\theta_i(H)$ induced by $H|y$. See Sorensen and Gianola (2002), page 552, for more details concerning the Rao-Blackwell estimator.

12. Page 361,
CHANGE:
"In terms of $p-$values, a threshold of 0.05 guarantees that the expected number of false positives over the whole family of m tests is less than or equal to $0.05\,m$, far too large for modern genomewide studies."
TO:
"In terms of $p-$values, a threshold of 0.05 guarantees that the expected number of false positives over the whole family of m tests is $0.05\,m$, far too large for modern genomewide studies."

13. Page 361,
CHANGE:
"Then for the whole family of m tests, when all the m hypotheses are true nulls, the expected number of false positives is less than or equal to $m(\alpha/m) = \alpha$."
TO:
"Then for the whole family of m tests, when all the m hypotheses are true nulls, the expected number of false positives is equal to $m(\alpha/m) = \alpha$."

14. Page 363,
CHANGE:
Then the expected number of false discoveries over the m tests is

$$\text{E}(V) = \text{E}\left(\sum\nolimits_{i\in M_0} V_i\right) = \sum\nolimits_{i\in M_0}\text{E}(V_i) \leq m_0\alpha_B.$$

TO:
Then, for continuously distributed tests statistics, the expected number of false discoveries over the m tests is

$$\mathrm{E}\,(V) = \mathrm{E}\left(\sum\nolimits_{i\in M_0} V_i\right) = \sum\nolimits_{i\in M_0} \mathrm{E}\,(V_i) = m_0\alpha_B.$$

15. Page 369,
CHANGE:
"This is in contrast with the meaning attached to the $p-$value for $i = 19$ equal to 0.02845: in conceptual replications of the experiment where a total of m hypotheses are tested, the expected number of false discoveries among the hypotheses rejected at the cutoff 0.02845 or smaller is $0.02845 \times m$ or smaller."
TO:
"This is in contrast with the meaning attached to the $p-$value for $i = 19$ equal to 0.02845: in conceptual replications of the experiment where a total of m hypotheses are tested of which m_0 are true nulls, for continuously distributed test statistics, the expected number of false discoveries is $0.02845\, m_0$."

16. Page 373, paragraph above Eq. (8.23), defines the number of false discoveries $V(A)$, given that k hypotheses are rejected, as binomially distributed
$Bi(k, \Pr(H = 0|z \in A)$. Another definition can be found on page 376, paragraph above Eq. (8.33), concerning the expected number of false discoveries. Both definitions may benefit from a little explanation that is provided below.

Background

Let the binary random variable $V_i = 1$ if hypothesis i is rejected and if hypothesis i is a true null; $V_i = 0$, otherwise. Using the notation on page 363, we can write
$V_i = I(H_i = 0, R_i = 1)$, where $R_i = 1$ if $z_i \in A$ and z_i is the observed value of the scalar random variable Z_i. Treating the hypotheses as random, it follows that V_i is a Bernoulli random variable with probability $\Pr(H_i = 0, Z_i \in A)$; that is $V_i \sim Br(\Pr(H_i = 0, Z_i \in A)$.

$$V_i \sim Br(\Pr(H_i = 0, Z_i \in A).$$

- **Page 373**. Consider the binary random variable $V_i|R_i = 1$ that takes the value 1 if $H_i = 0$ and 0 otherwise. Then $[V_i|R_i = 1] \sim Br(\Pr(H_i = 0|Z_i \in A)$, a Bernoulli distributed random variable. If in a series of independent trials the total number of rejections is $R = k$, among these k rejections, the total number of false discoveries is binomially distributed:

$$[V|R = k] \sim Bi(k, \Pr(H = 0|Z \in A).$$

The expected number of false discoveries among the k rejections is

$$\mathrm{E}(V|R=k) = k\ \Pr(H=0|Z\in A).$$

The subscript i has been omitted because $\Pr(H_i=0|Z_i\in A)$ is the same for each i.

- **Page 376**. Consider an experiment where m independent hypotheses are tested. Given the definition of V_i and the independence assumption, the total number of false discoveries, $V=\sum_{i=1}^{m} V_i$, is then binomially distributed

$$V \sim Bi(m, \Pr(H=0, Z\in A)),$$

with expected value

$$\mathrm{E}(V) = m\ \Pr(H=0, Z\in A)) = m\ \Pr(Z\in A|H=0)\pi_0,$$

where $\pi_0 = \Pr(H=0)$, the prior probability of a true null.

Note that: (i) hypotheses H_i are treated as random (with a common prior π_0), (ii) random vectors (H_i, Z_i), $i=1,2,\ldots,m$, are independently distributed, (iii) the rejection region A is data-independent and fixed in advance.

17. Page 382, CHANGE

$$\frac{\widehat{\mathrm{E}}\left(V|r, \widehat{\Pr}\left(\delta_i=0|y\right)\right)}{R} = \frac{\sum_{i=1}^{m} \widehat{\Pr}\left(\delta_i=0|y\right) r_i}{R}, \tag{8.42}$$

TO:

$$\frac{\widehat{\mathrm{E}}\left(V|r\right)}{R} = \frac{\sum_{i=1}^{m} \widehat{\Pr}\left(\delta_i=0|y\right) r_i}{R}, \tag{8.42}$$

18. Page 382, REPLACE THE CONTENT OF THE SUBSECTION:
"**Posterior Distribution of False Discovery Rate**"
WITH:
The following approach provides a Monte Carlo estimator of the complete posterior distribution of the fully Bayesian FDP (The *false discovery proportion*, FDP, is equal to the number of false discoveries, V, divided by the total number of discoveries, R; FDP $= V/R$). This distribution reflects posterior uncertainty of the parameters of the Bayesian model that propagates on to the FDP. The protocol is as follows. First execute the McMC sampler and store the draws $\delta_i^{[j]}$ at each cycle, where i refers to the marker genotype and j to the McMC cycle. On completion choose a discovery set; an informal approach could be based on selecting the marker genotypes whose Monte Carlo estimates of the posterior probability of non-zero effects is larger than a chosen value $\varphi^\star$. The resulting discovery set includes R marker genotypes. Then,

from the stored draws $\delta_i^{[j]}$ *belonging to the discovery set*, compute at each McMC cycle the false discovery proportion for the discovery set. For example, at round j compute

$$Q^{[j]} = \frac{\sum_{i=1}^{m} \left(1 - \delta_i^{[j]}\right) r_i}{R},$$
$$Q^{[j]} \in \left\{0, \frac{1}{R}, \frac{2}{R}, \ldots, 1\right\}, \quad j = 1, 2, \ldots, T, \tag{8.43}$$

where m is the number of markers and T is the number of samples. The numerator in the first line counts the number of indices i for which $\delta_i^{[j]} = 0$ and $r_i = 1$ in the jth iteration of the McMC chain. The quantity $Q^{[j]}$ is the jth McMC draw from the discrete posterior distribution of the Bayesian FDP for the discovery set chosen. Importantly, the discovery set is chosen only once and used repeatedly across all the draws of the McMC chain. With enough McMC samples the discreteness of the distribution $[Q \mid y, r]$ should become visible.

The MC (ordinary) Bayesian estimator of FDR is obtained by averaging over $Q^{[j]}$:

$$\hat{\mathrm{E}}(Q|y, r) = \frac{1}{T} \sum_{j=1}^{T} Q^{[j]},$$

that is an alternative way of expressing (8.42).

The two-component normal mixture model implemented in the example below uses rules of the type $r_i = I\left(y_i \geq y^\star\right)$ or $r_i = I\left(\hat{\varphi}_i \geq \varphi^\star\right)$.

A recent McMC implementation of the Bayesian FDR was described by de los Campos et al (2022)

Comment The purpose of this comment is to provide insight into the posterior variance of the false discovery proportion (FDP) and to show that the approach described in the book's current editions underestimates this variance.

A number of quantities must be defined. Let,

- $$\theta_i(H) = \Pr(\delta_i = 1 \mid y, H) = \mathrm{E}(\delta_i \mid y, H),$$

  where **here**, H contains all the parameters of the spike and slab model except the **complete** vector δ. In contrast to the single-marker Gibbs updates, where the conditioning vector excludes only the indicator being updated, we now exclude the complete vector δ, since the FDP depends simultaneously on all inclusion indicators in the discovery set. This simplifies the algebra.

- $$\delta_i \mid y, H \sim \text{Bernoulli}(\theta_i(H)).$$

- The number of false discoveries is

$$V = \sum_{i \in D} (1 - \delta_i),$$

where D is the discovery set containing $R > 0$ marker genotypes.

- The false discovery proportion is

$$Q = \frac{V}{R}.$$

The conditional expectation of the FDP is

$$\begin{aligned} \mathrm{E}(Q \mid y, H, D) &= \frac{1}{R} \sum_{i \in D} \mathrm{E}(1 - \delta_i \mid y, H) \\ &= \frac{1}{R} \sum_{i \in D} \left[1 - \theta_i(H)\right]. \end{aligned}$$

The posterior variance of the FDP, marginalised over $H|y$ is

$$\mathrm{Var}(Q \mid y, D) = \mathrm{E}_{H|y} \left[\mathrm{Var}\left(Q \mid y, H, D\right)\right] + \mathrm{Var}_{H|y} \left[\mathrm{E}(Q \mid y, H, D)\right].$$

The first term on the right-hand side accounts for the conditional posterior variation of the inclusion indicators δ_i, given H, including any conditional dependence among them. The second term reflects variation in the conditional expectation of the FDP induced by posterior uncertainty about the parameters in H. Consequently, using only the variance of the conditional expectation of the FDP (as in the book's current editions) underestimates the posterior variance of the FDP by the nonnegative quantity represented by the first term.